



Cambridge IGCSE™

MATHEMATICS

0580/22

Paper 2 (Extended)

May/June 2021

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **7** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

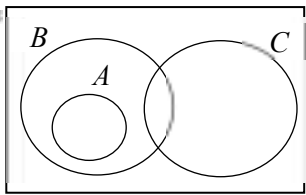
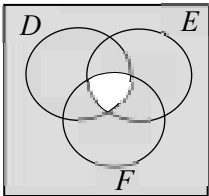
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Question	Answer	Marks	Partial Marks						
1(a)	$\frac{3}{10}$ oe	1							
1(b)	35	1							
2	0.4 or $\frac{2}{5}$	1							
3	<table border="1"><tr><td>Mode</td><td>16</td></tr><tr><td>Median</td><td>11</td></tr><tr><td>Range</td><td>17</td></tr></table>	Mode	16	Median	11	Range	17	3	B1 for each
Mode	16								
Median	11								
Range	17								
4	$k - 1$	1							
5(a)	It is not possible to tell if there is correlation as there are not enough points.	1							
5(b)	C	1							
6	Accurate construction of rhombus with sides 6.5 cm and correct construction arcs.	2	B1 for accurate diagram with no/wrong arcs or for one triangle (6.5 cm, 6.5 cm, 8 cm) correctly constructed with correct arcs or for four correct arcs						
7(a)	$\frac{5}{97}$	2	B1 for each						
7(b)	$\sqrt{7}$	1							
8	$[\pm] 7.5$ oe	2	M1 for $5.625 = \frac{b^2}{2 \times 5}$ or better						
9	$\frac{2}{3} \times \frac{7}{10}$ or $\frac{14}{21} \div \frac{30}{21}$ oe with common denominator	M2	B1 for $\frac{10}{7}$ oe or M1 for $\frac{2}{3} \times \text{their } \frac{7}{10}$						
	$\frac{7}{15}$ cao	A1							
10(a)	6.54×10^{-3}	1							
10(b)	99	1							
11	$\frac{4}{99}$ cao	1							
12(a)(i)	Even square numbers oe	1							

Question	Answer	Marks	Partial Marks
12(a)(ii)		1	
12(b)		1	
13	68	3	M1 for correctly identifying 90° angle soi or $DAC / DCA = 68$ M1 for [obtuse angle] AOC identified as $2x$ soi or $x = \text{their } DAC / DCA$
14	456 or 456.4...	4	M2 for $\frac{18.2}{\tan 62}$ oe or M1 for $\tan 62 = \frac{18.2}{x}$ oe M1 for $\frac{1}{2}((\text{their trapezium base}) + 15.4) \times 18.2$ oe
15	Congruent SAS Congruent SSS Not congruent None	3	B1 for each correct row
16(a)	8.94 or 8.944...	3	M2 for $\sqrt{(9-5)^2 + (-1-7)^2}$ oe or M1 for $(9-5)^2 + (-1-7)^2$ oe
16(b)	$y = -2x + 17$ oe final answer	3	B2 for answer $-2x + 17$ OR M1 for $\frac{-1-7}{9-5}$ oe M1 for correct substitution of (5, 7) or (9, -1) into $y = \text{their } mx + c$ oe
17	$-\frac{3}{4}$ or -0.75	2	M1 for $y = \frac{4x-5}{3}$ or better or for $\frac{-1}{\text{their gradient}}$

Question	Answer	Marks	Partial Marks
18	$[x =] -2.1$ oe	4	M3 for $x^2 + 10x = x^2 - 21$ or better OR M1 for $(x + 1 + 4)^2 - 25$ or better M1 for $x^2 - 25 + 4$ or better If 0 scored SC1 for answer $-\frac{11}{6}$ oe
19(a)	77.3 or 77.32 to 77.33...	3	M2 for $\frac{360-60}{360} \times \pi \times 12.4 \times 2$ oe [$\pm n \times 12.4$] or M1 for angle 60° or 300° soi or for $\frac{k}{360} \times \pi \times 12.4 \times 2$ oe [$\pm n \times 12.4$]
19(b)	5.17 or 5.172 to 5.173...	3	M2 for $\frac{74.5}{\pi} \times \frac{360}{360-41} = r^2$ oe or better or M1 for $74.5 = \frac{360-41}{360} \times \pi r^2$ oe or for $\sqrt{\frac{74.5}{\pi} \times \frac{360}{k}}$ oe
20	$2x^3 + 7x^2 - 7x - 30$ final answer	3	B2 for unsimplified expansion with at most one error or for simplified four-term expression of correct form with three terms correct or B1 for correct expansion of two brackets with at least three terms out of four correct
21(a)	$[F =] \frac{108}{d^2}$ final answer	2	M1 for $F = \frac{k}{d^2}$ oe or better
21(b)	$[n =] \frac{1}{4}$ or 0.25	1	
22	$\frac{2x+3}{3x}$ final answer	4	B2 for $(x-4)(2x+3)$ or B1 for $(x+a)(2x+b)$ where $ab = -12$ or $2a + b = -5$ or $x(2x+3) - 4(2x+3)$ or $2x(x-4) + 3(x-4)$ B1 for $3x(x-4)$
23	48.6 or 48.59... and 131.4 or 131.4...	2	B1 for each If 0 scored SC1 for two answers with a sum of 180°

Question	Answer	Marks	Partial Marks
24	$x = 3, x = -3$ nfww	5	M2 for $x + 9 + 9(x + 1) = (x + 1)(x + 9)$ oe or better or M1 for $x + 9 + 9(x + 1)$ or $(x + 1)(x + 9)$ oe or better B1 for $x^2 + x + 9x + 9$ seen M1 dep for $[0 =]x^2 - 9$ oe simplified or better